ECE 287 Project Proposal

Project Title: The Legend of Zelda: FPGA Adventure Game

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1. Project Overview

For our Digital System Design project, we propose designing and implementing a simplified recreation of the opening section of the original *The Legend of Zelda* (1986) for the NES.

The project will use the Altera DE2 FPGA development board to implement a playable adventure game entirely through digital hardware logic. Rather than emulating the original NES processor or using a traditional software-based game engine, we will design the game's control systems, graphics rendering, character movement, combat mechanics, and enemy behavior using Verilog HDL.

One of the main features of our project will be the use of an original Nintendo NES controller as the player's input device. We will design a digital interface allowing the FPGA to communicate with the controller and interpret button presses.

The game will be displayed on an external monitor through the DE2 board's VGA output, creating an interactive experience inspired by the original NES game.

2. Project Objectives

Our primary objective is to demonstrate the practical application of digital system design concepts through the development of a functional video game.

The project will incorporate:

- **Finite State Machines (FSMs):** Control game states, player attacks, enemy behavior, and communication with the NES controller.
- **Sequential Logic:** Store player coordinates, health, equipment, controller inputs, and game progress.
- **Combinational Logic:** Implement collision detection, movement decisions, button decoding, and graphical output.
- **Arithmetic Operations:** Calculate player and enemy positions, health changes, and timing-related operations.

- **Counters and Clock Control:** Manage VGA output, gameplay update timing, animations, and NES controller communication.
- **Memory and Data Storage:** Store graphics, background tile maps, and character sprite data using FPGA memory resources.

3. Proposed Game Features

The game will recreate a small portion of the original Zelda experience, focusing on the beginning of the adventure.

The player will begin in an overworld area containing trees, obstacles, and an entrance to a cave. Inside the cave, the player will obtain a wooden sword that can be used to attack enemies.

The proposed game will include:

- Three connected overworld screens.
- One cave containing the wooden sword.
- Four-directional player movement.
- Collision detection with walls, trees, and other obstacles.
- Sword collection and directional attacks.
- At least one type of enemy with simple movement behavior.
- A health system consisting of three hearts.
- Player damage and enemy defeat mechanics.
- A game-over condition when the player loses all health.
- A victory condition after completing a designated objective.

The game will use tile-based pixel graphics inspired by the original NES style. The graphics will be generated using hardware-based rendering logic and stored graphical data.

The scope will be intentionally limited to ensure that the game can be completed, tested, and demonstrated within the available project time.

4. Original NES Controller Integration

A major component of this project will be interfacing an original Nintendo NES controller with the Altera DE2 board using a WiiClassic adapter or a Raspberry Pi Pico to connect the DE2 to the NES controller.

The original NES controller uses a shift-register-based communication system that allows its eight button states to be read sequentially.

We will design a Verilog module that communicates with the controller using three primary signals:

- **LATCH:** Captures the current states of the controller's buttons.
- **CLOCK:** Advances the controller's shift register to transmit each button state.
- **DATA:** Returns the button information to the FPGA one bit at a time.

A dedicated controller finite state machine will generate the appropriate timing signals, read the serial data, and convert it into an eight-bit representation of the controller's buttons.

The intended button assignments will be:

Controller Button	Game Function
Up	Move player upward
Down	Move player downward
Left	Move player left
Right	Move player right
A	Attack with sword
B	Reserved for additional actions
Start	Begin or pause game
Select	Optional menu or restart function

The controller will connect to the DE2 through its GPIO expansion header using an appropriate adapter.

Because the original NES controller was designed for 5 V operation and the FPGA GPIO uses 3.3 V logic, we will verify the controller's electrical requirements and use a safe power and logic-level interface. A level-shifting circuit will be included if necessary.

We intend to use a detachable adapter so that the original controller does not need to be permanently modified.

5. Hardware Implementation

The Altera DE2 board will serve as the central processing platform for the project.

The FPGA will be responsible for both game processing and graphics generation.

The following hardware components will be used:

Hardware	Purpose
Altera DE2 Cyclone II FPGA	Execute digital game logic
VGA output	Display game graphics
Original NES controller	Provide player input
DE2 GPIO header	Interface with NES controller
FPGA internal memory	Store graphics and level data
DE2 clock circuitry	Provide timing for the system
DE2 switches and LEDs	Debugging and testing
Raspberry Pi pico / WiiClassic adapter	Connecting the NES controller to the DE2

The VGA subsystem will generate a 640 × 480 video signal. The game will use a lower-resolution logical pixel grid, scaled to fit the display while maintaining the appearance of classic NES graphics.

The design will be separated into independent Verilog modules, including:

1. VGA timing and synchronization.
2. NES controller communication and button decoding.
3. Main game state machine.
4. Player movement and direction control.
5. Environment and collision detection.
6. Room selection and transitions.
7. Sword and combat mechanics.
8. Enemy movement and behavior.
9. Health and game progression.
10. Graphics and sprite rendering.

These modules will communicate through defined signals and registers to form the complete game.

6. Development and Testing Plan

We plan to follow an incremental development process, building and testing individual components before integrating them into the complete system.

Phase 1: NES Controller Interface

Develop the Verilog communication module for the original NES controller. Test each button by displaying its state using the DE2's onboard LEDs.

Phase 2: VGA Display

Implement VGA timing and synchronization. Verify that the board can display a stable image and generate simple graphics on an external monitor.

Phase 3: Player Movement

Display the player character and use the NES controller's directional buttons to move around a basic environment.

Phase 4: Environment and Room System

Implement tile-based backgrounds, collision detection, room transitions, and the cave containing the wooden sword.

Phase 5: Combat and Enemy Behavior

Add sword attacks, enemy movement, damage detection, health tracking, and enemy defeat mechanics.

Phase 6: Game Integration and Testing

Combine all modules into a complete playable game. Test gameplay from beginning to end, correct bugs, and prepare for the live demonstration.

Whenever possible, individual modules will be tested through simulation before hardware implementation.

Project progress, Verilog source files, testbenches, and documentation will be maintained in a GitHub repository.

7. Project Scope and Limitations

The goal of this project is to create a short but complete playable experience rather than recreating the entire original Zelda game.

The initial design will not include:

- The complete original overworld.
- Multiple dungeons or boss encounters.
- A complex inventory system.
- Multiple weapons or advanced combat mechanics.
- Sophisticated enemy artificial intelligence.
- Game saving or persistent progress.
- Continuous scrolling between locations.

If development progresses ahead of schedule, additional features may include character animations, sound effects, additional enemies, or a small overworld setting.

The project will prioritize reliable operation and complete functionality over the number of features implemented.

8. Expected Deliverables

Upon completion, the project will include:

- A functioning mini-Zelda adventure game running on the Altera DE2 FPGA.
- An original NES controller successfully interfaced with the FPGA.
- A VGA display showing the game environment, character, enemies, and health system.
- A GitHub repository containing all Verilog source files, testbenches, and relevant documentation.
- A technical report describing the design architecture, FSMs, digital logic implementation, testing procedures, and results.
- A live demonstration for Dr. Jamieson showing gameplay controlled using the original NES controller.

9. Expected Outcome

The completed project will allow a player to use an original Nintendo NES controller to navigate a mini Zelda adventure, defeat enemies, and complete a final objective.

The project will demonstrate how finite state machines, combinational logic, sequential logic, arithmetic operations, memory structures, and external hardware communication can be integrated into a functional digital system.

By combining FPGA-based game development with an authentic NES controller, we aim to produce a technically challenging, visually engaging, and well-documented project that demonstrates our understanding of digital system design.